电场不均匀. 磁场均匀

粒子运动方程. $m\frac{d\vec{v}}{dt} = q[\vec{E} + \vec{v}\times\vec{B}]$.

$$\begin{cases} m\frac{dv_x}{dt} = qE_x(x) + qBv_y \Rightarrow \dot{v}_x = \omega_c v_y + \frac{\omega_c}{B}E_x(x) \\ m\frac{dv_y}{dt} = -qBv_x \qquad\qquad \dot{v}_y = -\omega_c v_x. \\ m\frac{dv_z}{dt} = 0 \end{cases}$$

$$\ddot{v}_x = -\omega_c^2 \dot{v}_y + \frac{\omega_c}{B}\frac{dE_x(x)}{dt}.$$

$$\ddot{v}_y = -\omega_c^2 v_y - \omega_c^2 \frac{E_x(x)}{B}.$$

$E=0$ 有: $x = x_0 + r_L\sin(\omega_c t)$

$$\ddot{v}_y = -\omega_c^2 v_y - \omega_c^2 \frac{E_x}{B}\cos[k(x + r_L\sin\omega_c t)].$$

$\cos(kx)\cos(kr_L\sin\omega_c t) - \sin kx\sin kr_L\sin\omega_c t$.

$= \cos kx_0(1-\frac{1}{2}k^2r_L^2\sin^2\omega_c t) - \sin(kx_0)\, kr_L\sin\omega_c t$ (Taylor 展开).

$\ddot{\bar{v}}_y = -\omega_c^2\bar{v}_y - \omega_c^2\frac{E_0}{B}\cos(k(x_0 + r_L\sin\omega_c t)) = 0$.

$\cos\varepsilon = 1-\frac{1}{2}\varepsilon^2$

$\sin\varepsilon = \varepsilon$.

$$\bar{v}_y = -\frac{E_0\cos kx_0}{B}(1-\frac{1}{4}k^2r_L^2)$$

$$= -\frac{E_x(x)}{B}(1-\frac{1}{4}k^2r_c^2)$$

$$v_E = \frac{\vec{E}\times\vec{B}}{B^2}\left(1-\frac{1}{4}k^2r_c^2\right). \qquad kr_L \ll 1.$$

对变化电场，

$$v_E = \left(1+\frac{1}{4}r_L^2\nabla^2\right)\frac{\vec{E}\times\vec{B}}{B^2}$$

离子与电子半径不相同，漂移不稳定.

$\therefore$ $v_E \Rightarrow$ 分离 $\Rightarrow E\uparrow \Rightarrow$ 等离子体不稳定.